

Building the prediction machine

Model selection

Design a model that matches the research question and fulfills domain-specific requirements, for example:

- control for confounders
- capturing aspects of the data-generating process
- goodness of fit, within-sample or out-of-sample
- ...

Model fitting

Fit the statistical model within your preferred framework

- Bayesian model
- Frequentist model
- machine learning model

In the “standard” workflow, we study the estimated coefficients directly.

In the “prediction machines” workflow, we use the coefficients to compute target quantities.

Querying the prediction machine

Target quantity selection

Select one or multiple target quantities to answer the research question.

Predictions

What is the expected outcome for different types of people/units?

Comparisons and slopes

How does the expected outcome change when one or more predictors change?

Averaging (or not)

Decide for which unites to calculate target quantities.

- unit-level estimates
 - at observed values
 - at the mean
 - at representative values
 - ...
- average estimates
 - subgroup averages

Scale and contrast

Decide on which scale to calculate target quantities. The available options depend on the model type:

- link
- response
- ...

For counterfactual comparisons, decide how the predictions are contrasted:

- difference
- ratio
- ...

Testing the results

Statistical inference

- Quantify uncertainty (delta method, bootstrap, simulation, Bayesian posterior)
- Conduct null hypothesis or equivalence tests on coefficients or target quantities
- Test interactions
 - Is the expected outcome higher in group A than in group B?
 - Is the effect or association stronger for some individuals?
 - ...